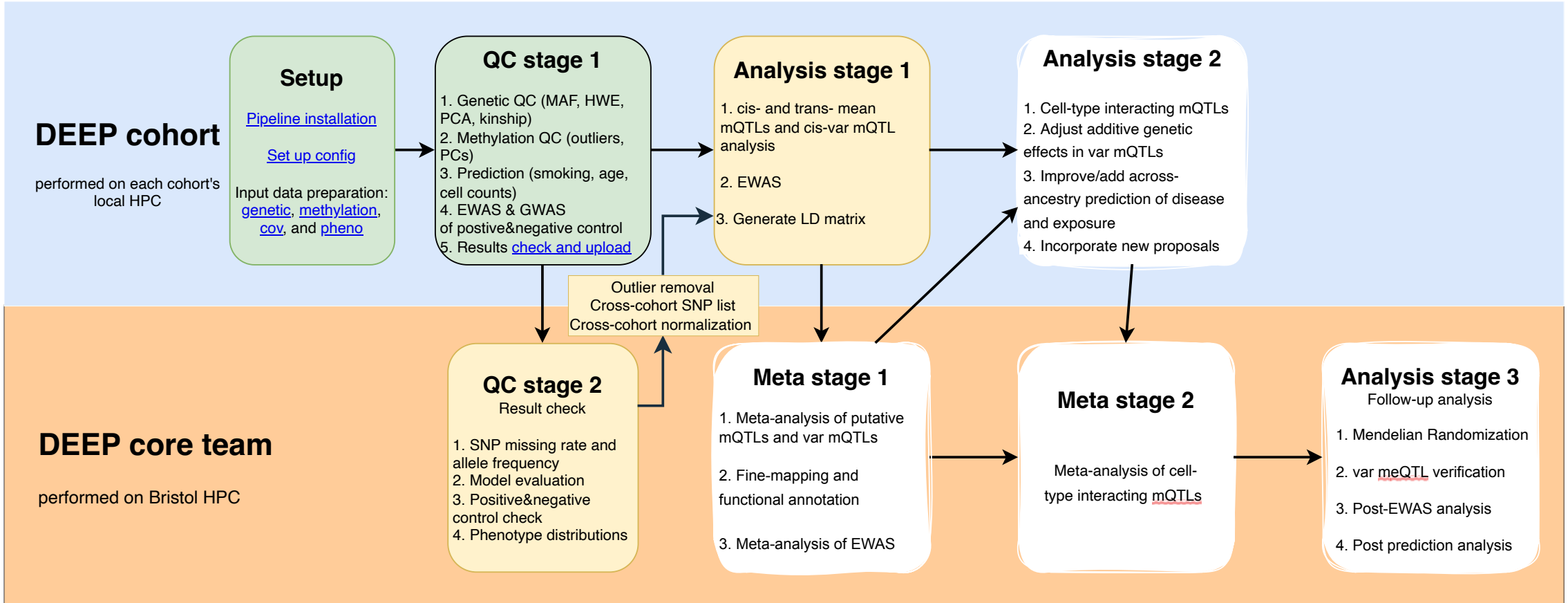

DEEP cohort
performed on each cohort's local HPC
Setup
Pipeline installation
Set up config
Input data preparation: genetic, methylation, cov, and pheno
QC stage 1
1. Genetic QC (MAF, HWE, PCA, kinship)
2. Methylation QC (outliers, PCs)
3. Prediction (smoking, age, cell counts)
4. EWAS & GWAS of postive&negative control
5. Results check and upload
Analysis stage 1
1. cis- and trans- mean mQTLs and cis-var mQTL analysis
2. EWAS
3. Generate LD matrix
Analysis stage 2
1. Cell-type interacting mQTLs
2. Adjust additive genetic effects in var mQTLs
3. Improve/add across-ancestry prediction of disease and exposure
4. Incorporate new proposals
Outlier removal
Cross-cohort SNP list
Cross-cohort normalization
DEEP core team
performed on Bristol HPC
QC stage 2
Result check
1. SNP missing rate and allele frequency
2. Model evaluation
3. Positive&negative control check
4. Phenotype distributions
Meta stage 1
1. Meta-analysis of putative mQTLs and var mQTLs
2. Fine-mapping and functional annotation
3. Meta-analysis of EWAS
Meta stage 2
Meta-analysis of cell-type interacting mQTLs
Analysis stage 3
Follow-up analysis
1. Mendelian Randomization
2. var meQTL verification
3. Post-EWAS analysis
4. Post prediction analysis